

PART I: Multiple Choice [24 pt]

Choose the best answer from the given alternatives and give your answer on the space provided in the answer sheet. [1.5 pt each]

1. Of the given alternative which one mismatched with its data unit in the case of TCP/IPv-2 reference model **(CLO - 3)**
 - A. Application Layer : Message
 - B. Transport Layer: Segment
 - C. Data-Link Layer: Ethernet Frame
 - D. Network Layer: Frame
 - E. All except D
2. Why is the User datagram protocol known to be connectionless, because **(CLO-4)**
 - A. Every segment are treated independently by transport layer
 - B. It sends the pieces of information very quickly
 - C. It disseminate information in the way ordered
 - D. It passes the data as a stream of related segment
3. Assume a source address 2001:0db8:85a3::7225 and the destination ip address 10.45.58.120. Which IPv-6 reserved address is responsible for mapping the source address to the destination **(CLO-4)**
 - A. Link-local Address
 - B. Compatible Address
 - C. Mapped Ip address
 - D. IPv-6 Converter
4. Identify a modulation scheme which utilize two different amplitudes of the carrier frequency to represent two binary level/values **(CLO-2)**
 - A. Amplitude Modulation
 - B. Amplitude Shifting keying
 - C. Quadratic shifting Keying
 - D. Binary-amplitude shifting Keying
5. A multiple access protocol which don't allow station not to continuously sense, when the channel is busy, rather it waits a random amount of time and repeat the algorithm is **(CLO-2)**
 - A. Persistent CSMA
 - B. Non-Persistent CSMA
 - C. Reservation
 - D. All
6. One explains synchronization service in session layer, is technique **(CLO-3)**
 - A. which assign time for nodes during communication
 - B. Prevents a nodes for sending or receiving pieces of information simultaneously
 - C. That lets the nodes resume the communication after some sort of failure occurred
 - D. It prevent simultaneously critical operation
7. The OSI Reference Model provides for all of the following except **(CLO-1)**
 - A. Defines the process for connecting two layers together, promoting interoperability between vendors
 - B. Allows vendors to compartmentalize their design efforts to fit a modular design, which eases implementations and simplifies troubleshooting
 - C. Separates a complex function into simpler components
 - D. Provides only theoretical help network administrators understand the communication process used between networking components

8. Discover an elements that are used to provide a reliable connection **(CLO-3)**
- A. Ready/not ready signals
 - B. Windows
 - C. Sequence numbers and acknowledgments
 - D. Ready/not ready signals and windowing
9. Which of the following is not a feature of a protocol that is considered to match OSI Layer 4 ? **(CLO-3)**
- A. Error recovery
 - B. Segmenting of application data
 - C. Flow control
 - D. Conversion from binary to ASCII
10. A type of bridge, which is responsible for converting data format in the data-link layer. **(CLO-1)**
- A. Converter Bridge
 - B. Translational Bridge
 - C. Source Routing Bridge
 - D. Transparent Bridge
11. Which TDM variant where the output frame collects data from the input frame until it is full, not leaving an empty slot **(CLO-2)**
- A. Synchronous TDM
 - B. Output TDM
 - C. Asynchronous TDM
 - D. All except B
12. What do we call a transmission impairment occur as a result of the effect of one wire on the adjacent wire **(CLO-1)**
- A. Thermal noise
 - B. Crosstalk
 - C. Impulse noise
 - D. Induced noise
13. A typical type of cables which commonly used in Ethernet Local area Network: **(CLO-2)**
- A. Copper Wire
 - B. Unshielded twisted-pair (UTP)
 - B. Shielded twisted-pair (STP)
 - D. Optical Fiber cable
14. Supposedly, you are testing connectivity with the ping command and getting a “request timeout” or “destination unreachable” message in reply. At what layer of the OSI Reference Model does this indicate a problem? **(CLO-3)**
- A. Transport Layer
 - B. Data link Layer
 - C. Network Layer
 - D. Application Layer
15. Quesiton ???????
16. Question ???????

Part II: Short Answer [16pt]

Clearly and neatly answer the following the questions accordingly

1. Explain the key services in a session and presentation layer of OSI model. [3 pt] (CLO-4)
2. Compare and contrast IPv-4 and IPv-6 in terms of size, representation and security. [3 pt] (CLO-4)

3. Question

4. Question

5. Question

Part III: Short Answer [10pt]

Clearly and neatly answer the following the questions accordingly

1. Assume a PDU “**GoodLuck**” prepared at the Application layer. Apply Non-Return-to-Zero Inverted line encoding scheme to represent pieces of the PDU separately at the physical layer. [4 pt] (CLO-2)
2. Suppose Ethio-Google enterprise has requested you to analyze the network topology below and design logical addressing. For future scalability the company gave you 172.20.12.0 as its default network with its default subnet mask. Based on the topology determine [6pt] (CLO-4)
 - A. Determine the Bit-to borrow for each NET
 - B. The subnet mask of each NET
 - C. The network address of each NET
 - D. The address space of each NET
 - E. The broadcast address of subnetwork

